



GROWTH & GH AXIS

Ipamorelin

Ipamorelin – 10 mg per vial – 10 vials per box

FOR RESEARCH USE ONLY (RUO). NOT FOR HUMAN OR VETERINARY USE.

IDENTIFICATION

CHEMICAL NAME	Ipamorelin
SYNONYMS / ALIASES	NNC 26-0161; selective growth hormone secretagogue
CAS NUMBER	170851-70-4
MOLECULAR FORMULA	C38H49N9O5
MOLECULAR WEIGHT	711.85 g/mol
AMINO ACID SEQUENCE	H-Aib-His-D-2Nal-D-Phe-Lys-NH2 (5-residue pseudo-peptide; Aib = alpha-aminoisobutyric acid; D-2Nal = D-2-naphthylalanine; C-terminal amide)
SOURCE	Synthetic (solid-phase peptide synthesis)

PHYSICAL PROPERTIES

APPEARANCE	Lyophilized white to off-white powder
PURITY SPECIFICATION	>=99% by HPLC (confirm against batch COA)
CONTAINER	10 mg per vial, sealed glass vial, rubber-stoppered, aluminum crimp

STORAGE & HANDLING

UNRECONSTITUTED STORAGE	-20 C, protect from light and moisture
RECONSTITUTED STORAGE	2-8 C, stable approximately 28 days in bacteriostatic water
SUGGESTED RECONSTITUTION SOLVENT	Bacteriostatic water or sterile water (research use only)
SHELF LIFE (UNRECONSTITUTED, SEALED)	24 months at -20 C
STABILITY NOTES	The Aib and D-amino acid substitutions confer enzymatic stability. Avoid repeated freeze-thaw cycles.

STRUCTURAL NOTES

Ipamorelin is a synthetic 5-residue pentapeptide containing two non-natural amino acids: alpha-aminoisobutyric acid (Aib) at the N-terminus and D-2-naphthylalanine (D-2Nal) at position 3, plus a D-phenylalanine at position 4. The C-terminus is amidated. Structurally it is classified as a ghrelin / growth hormone secretagogue receptor (GHS-R1a) ligand. The substitution pattern was designed to confer high in vitro selectivity for the GHS-R1a receptor relative to other secretagogues.

SELECTED PEER-REVIEWED REFERENCES

1. Raun K. et al. - *Ipamorelin, the first selective growth hormone secretagogue* - European Journal of Endocrinology, 1998 - DOI: 10.1530/eje.0.1390552
2. Andersen N.B. et al. - *The growth hormone secretagogue ipamorelin counteracts glucocorticoid-induced loss of cortical bone in rats* - Journal of Bone and Mineral Research, 2001 - DOI: 10.1359/jbmr.2001.16.7.1280
3. Gobburu J.V. et al. - *Pharmacokinetic-pharmacodynamic modeling of ipamorelin, a growth hormone releasing peptide, in human volunteers* - Pharmaceutical Research, 1999 - DOI: 10.1023/A:1018826704879

References are provided for research background only. Cited publications do not constitute claims about the safety, efficacy, or intended use of this product.

COMPLIANCE DISCLAIMER

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Batch-specific identity and purity are documented in the Certificate of Analysis (COA) issued for the lot.